NUMBER RACK (REKENREK) ARITHMETIC STUDIO

Dual-Row 20-Bead 5-and-10 Grouping & Subitizing Frame

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · 5·Primes · 6·Algebra · 7·Probability · 8·Volvelle · 9·Quest · **10·Rack** · 11·SC-32

CARD 11 / 11 • RACK-01

MATHEMATICAL REFERENCE

1. WHAT IS THIS MANIPULATIVE?

5-AND-10 GROUPING. SUBITIZE DUAL-ROW ADDITION.

Counting by ones is slow and error-prone.
The Rekenrek organizes 20 beads into 5-anchors:
Row 1: 5 Red + 5 White • Row 2: 5 Red + 5 White
Sliding beads in groups turns math into subitizing!

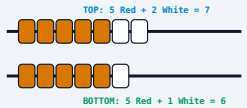
CALIBRATED 102.0 MM SLIDER DATUM

- Beads: 10 Red + 10 White = 20 Beads (14.0 mm ⌀)
- Slider Rods: 2x 102.0 mm Octagonal Rods
- Upright Towers: 2x 49.0 mm Snap-Fit Towers (7U)
- Connector Spec: 1.85mm base plug & stacking socket

DUAL-SLIDER COMPARISON STAGE LID

Snap upright towers into lid for a counting frame.
Slide beads left to add, right to read sums!

5-AND-10 GROUPING: 7 + 6 = (5+2) + (5+1) = 13



DOUBLES-PLUS-ONE

- Red Beads: 5 + 5 = 10 (I
- White Beads: 2 + 1 = 3 U

Sum = 10 + 3 = 13

Zero finger counting!

2. KIT PARTS INVENTORY (27 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Red Beads	10x	14.0 mm ⌀ Miniblock Counter	Crimson Red
White Beads	10x	14.0 mm ⌀ Miniblock Counter	Classic White
Slider Rods	2x	102.0 mm Octagonal Rod	Matte Charcoal
Upright Towers	2x	49.0 mm (7U) Mounting Tower	Matte Charcoal
Storage Tray	1x	Internal Nesting Tray	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal
Stage Lid	1x	Dual-Slider Snapfinity Lid	Matte Charcoal

TOTAL: 20 Beads + 4 Frame Parts + Tray + Storage Bin + Lid = 27 Parts

DUAL-ROW 5-COLOR STRUCTURE

- 5 Red + 5 White per row: Instant 5-anchor visual subitizing.
- Both Rows Active: Models doubles, near-doubles, and bridging.
- Zero-Rattle Carry: Beads slide into nested tray slots.
- High Contrast: Crisp color separation across rows.

3. WHY KEEP THIS REFERENCE CARD?

✓ NO LOST PIECE LEFT BEHIND

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Slide all 10 red beads to the left and all 10 white beads to the right on both rods. Place rods and towers into the internal nesting tray. Any missing bead leaves an immediately visible gap. Lid latches securely.

✓ HANDS-ON TUTORIAL ON REVERSE

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

DUTCH REKENREK DUAL-ANCHOR THEOREM

Dual 5-and-10 grouping bridges perceptual subitizing directly into formal arithmetic operations.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 11 OF 11 • DOMAIN: 5-AND-10 GROUPING & DUAL-ROW ARITHMETIC • 20-BEAD REKENREK • 2 ROWS × 5 RED + 5 WHITE

LESSON GUIDE: DUAL-ROW BRIDGING & DOUBLES PLUS ONE

Step-by-Step Hands-On Tutorial: 7 + 6 = 13 via 5-and-5 Doubles

Grades K-3 Subitizing & Addition • Hands-On Classroom & Home STEM Studio Series

CARD 11 / 11 • RACK-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/number-rack/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.1.OA.C.6 (Addition Fluency)
- CCSS.MATH.CONTENT.2.OA.B.2 (Mental Math)
- TEKS 111.3.b.3.C (Compose/Decompose)

Grades K-3 Subitizing & Addition

MATHEMATICAL INQUIRY

When you slide 7 beads on top and 6 on bottom, how many red beads are on the left side altogether?

1 STEP 1: SLIDE 7 BEADS ON TOP ROW

Push all beads to the right side of the rack (resting zero).
Slide 7 beads to the left on the top rod in one push:
All 5 red beads + 2 white beads = 7 beads!
Notice: 5-anchor lets you see 7 instantly without counting.

2 STEP 2: SLIDE 6 BEADS ON BOTTOM ROW

Slide 6 beads to the left on the bottom rod in one push:
All 5 red beads + 1 white bead = 6 beads!
We want to find 7 + 6 without counting one-by-one.

3–4 STEPS 3 & 4: GROUP DOUBLES & REGROUP

3. GROUP RED BEADS: Look at the red beads on the left: 5 + 5 = 10 (Instant Ten)!
4. ADD WHITE BEADS: Look at the white beads: 2 + 1 = 3 units.
Total: 10 + 3 = 13! Therefore: 7 + 6 = 13!
Calculator Verification: Press Number Rack Subitizing Icon ([▣]) and run [7] [ENTER] [6] [+] → 13.0000.

5 CORE PRINCIPLE: DOUBLES MAKE MENTAL MATH EFFORTLESS

Instead of counting 6 fingers past 7, the Rekenrek decomposes 7 + 6 into (5 + 2) + (5 + 1). The two groups of 5 red beads instantly make 10, leaving 3 white beads. Addition becomes instantaneous visual pattern recognition rather than tedious memorization!

★ EXTENSION INVESTIGATION: SOLVE: 8 + 7 USING 5-AND-5

Slide 8 beads on top (5 red + 3 white) and 7 on bottom (5 red + 2 white). Group the 10 red beads first! What is the sum? (Scan QR for 3D animated solution!)

→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

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CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

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